



Abd Alrahman Ahmed Mousa Mohamed

Computer Science & Information ·
Software Developer

Phone	01029646085
Date of birth	6 / 6 / 2004
Age	22
Education	حاسبات و معلومات
GitHub	github.com/engboda6
Portfolio	engboda6.github.io/portf olio

My Work

1. Cosmic Observer

Windows astronomy desktop app —
3D sky, NASA data, AI tutor.

2. Vertical Farm Analysis

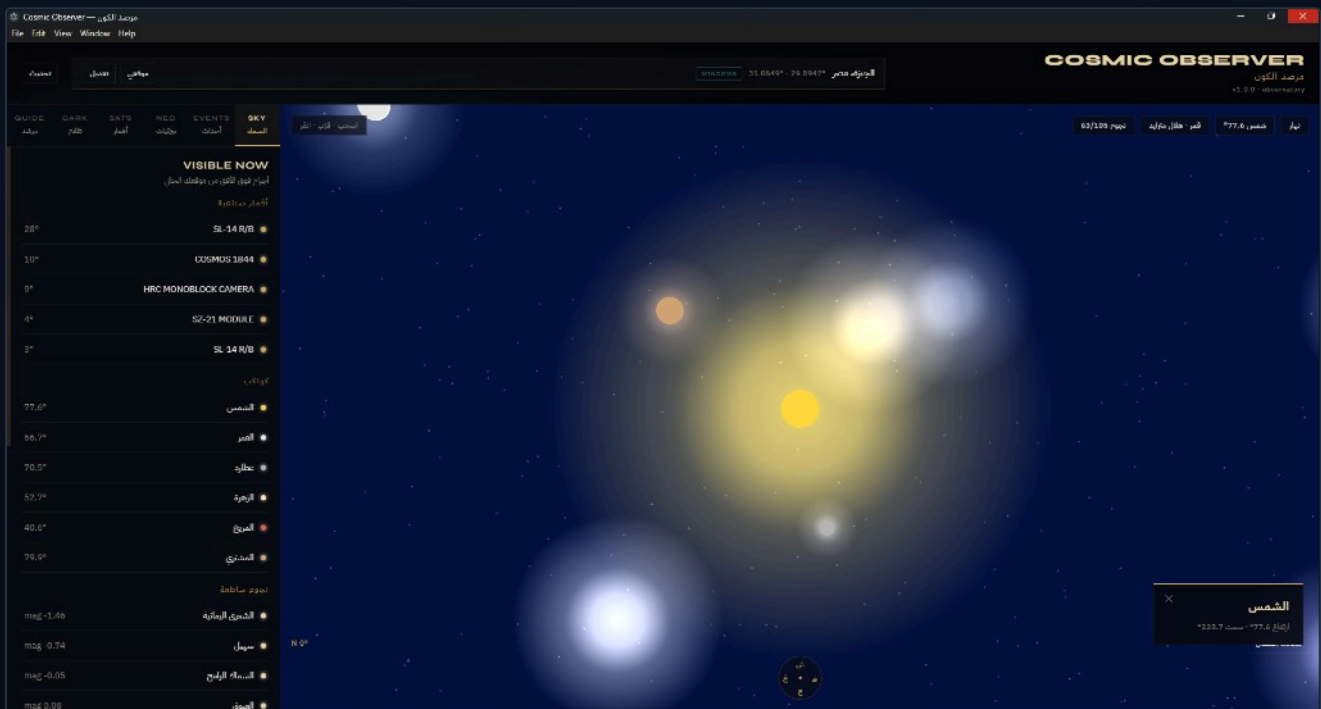
Python pipeline — farm telemetry to
charts & PDF report.

I build software that clients can see and test — desktop apps and data pipelines with real outputs: interactive demos, charts, and delivery-ready reports. This document presents my two main projects in detail.

PROJECT 01

Cosmic Observer — مرصد الكون

A Windows desktop astronomy application with an interactive 3D sky, near-Earth object tracking, orbital debris, a dark-matter lensing research module, and an AI astronomy tutor — packaged as a native **.exe**.



What you see

- Live 3D sky from GPS / manual location (Giza example)
- Visible satellites, planets, and bright stars
- Sun / Moon status, elevations, and sky events
- Tabs: Sky, Events, NEO, Sats, Dark, Guide
- Bilingual UI (Arabic + English)

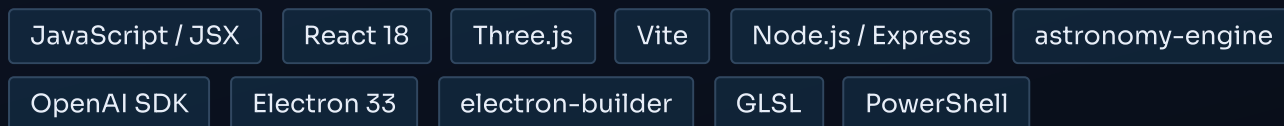
Repository

- GitHub: github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows .exe (NSIS + Portable)
- Demo video on portfolio site

Cosmic Observer — Technical Detail

Built end-to-end in JavaScript: React + Three.js for the UI and 3D sky, Node.js/Express for local APIs and orbital math, Electron for the Windows desktop shell.

Tech stack



Highlights

- **3D Sky** — stars, planets, Sun, Moon, Milky Way
- **Sky Events** — sunrise/sunset, moon phases, ISS passes
- **NEO Tracking** — NASA NeoWs with risk scoring
- **Orbital Debris** — CelesTrak TLE data
- **Dark Matter** — research-style lensing / probability maps
- **AI Tutor** — OpenAI chat (mock mode without a key)

Live data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS star clusters

Architecture

- `electron/` — main + preload
- `server/` — astronomy & data APIs
- `src/` — React UI + Three.js
- `scripts/` — build / run helpers

PROJECT 02

Vertical Farm Telemetry Analysis

A pure Python data analysis system for a vertical farm. It reads a real telemetry ledger with **600 production rows** and produces a clean dataset, analysis tables, charts, Excel exports, and a delivery-ready PDF report.

Pipeline (one command)

1 Import & clean — prepare_dataset.py

Reads `vertical_farm_biomass_telemetry_ledger.csv`, cleans columns, parses Stack / Rack / Row IDs, outputs an analysis-ready dataset.

2 Analysis — analysis.py

Nutrient decay acceleration matrix for 60 irrigation channels, weak-circulation channels, harvest-weight index per crop, best nutrition balance from top yield rows.

3 Charts — charts.py

EC distribution, circulation vs nutrient decay, matrix heatmap, harvest index plot.

4 Report — report.py

3-page PDF (Basant) with summary, matrix, and recommendations.

5 Full run — run_pipeline.py

Runs every step above with a single command.

Vertical Farm — Stack & Delivery

Language: **Python only**. Raw farm CSV → organized dataset + analysis tables + charts + PDF ready for portfolio / client delivery.

Libraries

- **pandas** — read, clean, aggregate data
- **numpy** — calculations & normalization
- **matplotlib** — charts + PDF generation
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis for 60 irrigation channels
- Charts (EC, heatmap, harvest index)
- Excel files
- 3-page PDF report

Contact

Abd Alrahman Ahmed Mousa Mohamed

Phone: 01029646085

Education: حاسبات و معلومات · Age 22 · Born 6/6/2004

GitHub: github.com/engboda6

Portfolio: engboda6.github.io/portfolio

Email: abdalrahmanahmedmousa@gmail.com

Live portfolio site includes Cosmic Observer screenshots, demo video, and full project pages for both works above.